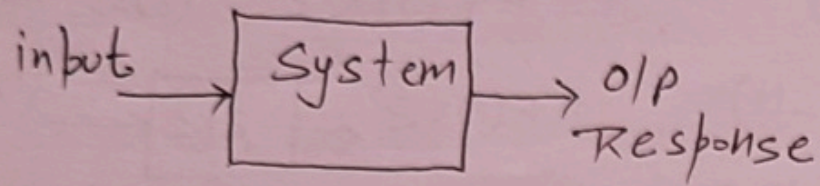
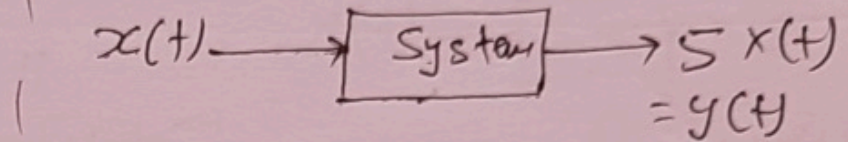


Types of System



Eg:- $y(t) = 5x(t)$



① linear & Non-linear system

* obey Homogeneity Principle

* obey Superposition Principle

$$x_1(t) \longrightarrow 5x_1(t) \checkmark$$

$$x_2(t) \longrightarrow 5x_2(t) \checkmark$$

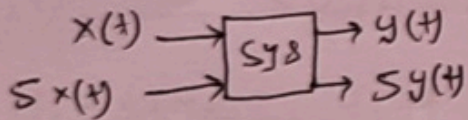
$$(x_1(t) + x_2(t)) \longrightarrow 5(x_1(t) + x_2(t))$$

$$\frac{5x_1(t) + 5x_2(t)}{=}$$

$$5x_1(t) + 5x_2(t)$$

linear system

$$\begin{array}{l|l} x(t) \longrightarrow y(t) & x_1(t) \longrightarrow y_1(t) \\ a x(t) \longrightarrow a y(t) & x_2(t) \longrightarrow y_2(t) \\ & x_1(t) + x_2(t) \longrightarrow y_1(t) + y_2(t) \end{array}$$

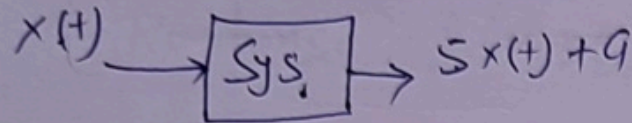


Note:- if one of the Property not followed then system is Non-linear.

Problems on linear & Non linear System

Non linear

① $y(t) = 5x(t) + 9$



$x(t) \rightarrow y(t)$
 $ax(t) \rightarrow ay(t)$

$ax(t) \rightarrow a(5x(t) + 9)$
 $\underline{5ax(t) + 9a}$

$ax(t) \rightarrow \boxed{\text{Sys.}} \rightarrow \underline{5ax(t) + 9}$
 $\downarrow$ Homogeneity $\times$

$x_1(t) \rightarrow y_1(t) = 5x_1(t) + 9$

$x_2(t) \rightarrow y_2(t) = 5x_2(t) + 9$

$(x_1(t) + x_2(t)) \rightarrow y_1(t) + y_2(t)$

$(x_1(t) + x_2(t)) \rightarrow 5(x_1(t) + x_2(t)) + 9$

$5x_1(t) + 9 + 5x_2(t) + 9$

$5x_1(t) + 5x_2(t) + 18 \neq 5x_1(t) + 5x_2(t) + 9$

Superposition $\times$

② $y(t) = tx(t)$

$x(t) \rightarrow tx(t)$

$ax(t) \rightarrow ay(t)$

$ax(t) \rightarrow \underline{t \cdot ax(t)} \checkmark$

$x_1(t) \rightarrow y_1(t) = tx_1(t)$

$x_2(t) \rightarrow y_2(t) = tx_2(t)$

$(x_1(t) + x_2(t)) \rightarrow y_1(t) + y_2(t)$
 $\underline{tx_1(t) + tx_2(t)}$

$(x_1(t) + x_2(t)) \rightarrow \boxed{\text{Sys.}}$

$\downarrow$
 $t(x_1(t) + x_2(t))$

$\underline{tx_1(t) + tx_2(t)}$
 Superposition $\checkmark$

linear $\checkmark$

③ $y(t) = e^{x(t)}$

$x(t) \rightarrow \boxed{\text{Sys.}} \rightarrow e^{x(t)}$

$ax(t) \rightarrow ay(t)$
 $\underline{ae^{x(t)}}$

$ax(t) \rightarrow \boxed{\text{Sys.}} \rightarrow \underline{e^{ax(t)}}$

$\frac{ay(t)}{ae^{x(t)}} \neq e^{ax(t)}$

System Non-linear

Causal & Non-Causal System

Causal System

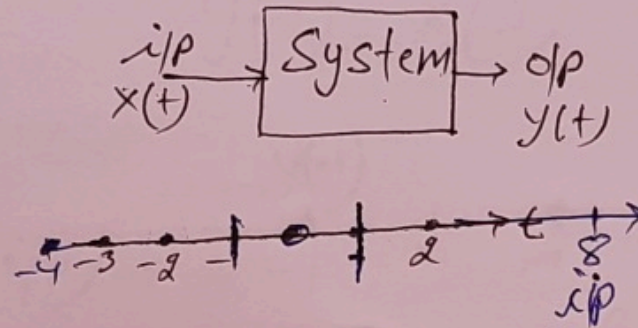
- * Response = $f(\text{Past, Present})$
input
- * Response = $o/p \neq f(\text{future value})$
of i/p

* Physically Verified

* Non-anticipatory

Eg:- $y(t) = e^{x(t-3)}$
 $y(1) = e^{x(-2)}$
 $y(-1) = e^{x(-4)}$ input $t = -2 \text{ sec}$

Causal System



Non-Causal System

- * Response = $f(\text{future value})$
of i/p
- * Physically Unrealizable
- * Anti-icipatory System

Eg:- $y(t) = \log(x(t+7))$
 $= \log(x(t+7))$
 $y(1) = \log(x(8))$

Non causal system

Problems on Causal & Non Causal System

①

$$y(t+3) = x(t+2) \cdot x(t-1)$$

$t=1$

$$y(4) = x(3) \cdot x(1)$$

$t=-1$

$$\underline{y(2)} = x(1) \cdot x(-2) \quad \checkmark$$

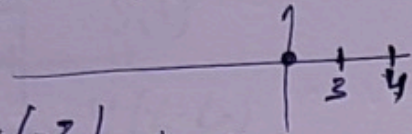
$t=10$

$$y(13) = \underline{x(12)} \cdot \underline{x(9)}$$

Past input

$o/p = f(\text{Past, Present i/p})$

Causal system



②

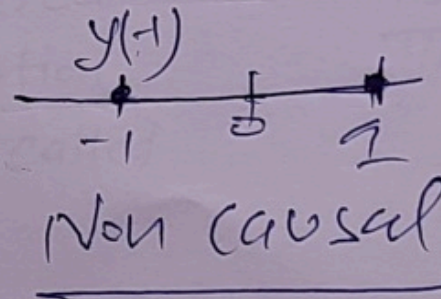
$$y(t) = x(-t)$$

1 sec

$$y(1) = x(-1) \quad \checkmark$$

-1 sec

$$y(-1) = x(1)$$



Non causal

③ $y(t) = x(\sin t)$

$$y(t) = x(\sin t)$$

$t=\pi$

$$y(\pi) = \underline{x(0)} \quad \checkmark$$

Past value

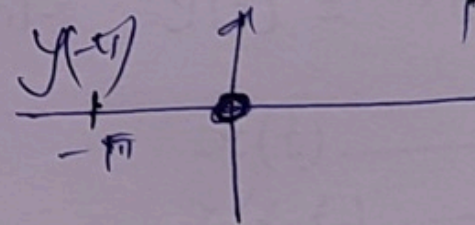
$t=-\pi$

$$y(-\pi) = x(\sin(-\pi))$$

$$= x(-\sin \pi)$$

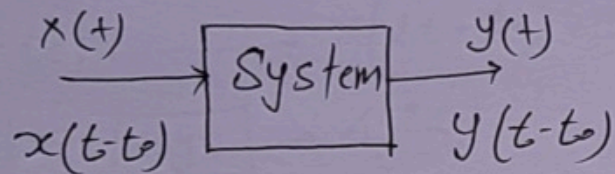
$$= \underline{x(0)}$$

Non causal system



Time Variant OR Time Invariant

Time invariant



Def:- if a system Generates Same Response for considered input irrespective of Application of time of input, system is called as Time invariant.

* Same Delay in input leads to Same Delay in Response.

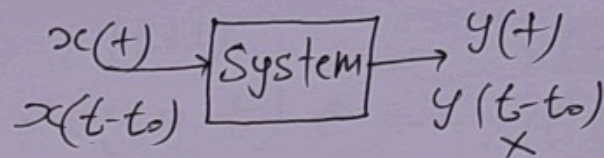
Steps

① Introduce delay of t_0 in input ^{o/p} in time.

② Replace $t \rightarrow t - t_0$ in given eqⁿ show system behaviour.

③ if ① = ② Time invariant otherwise T.V.

Time Variant



Def. if system generates diff. Response for same input when input is Applied at diff. Point of time, it is called Time variant system.

* Same Delay in input does not lead to same Delay in Response.

Eg:- $y(t) = x(-t)$

$$x(t) \longrightarrow x(-t)$$

$$x(t-t_0) \longrightarrow \underline{x(-t-t_0)} = y(t+t_0)$$

System behaviour

$$\underline{x(t-t_0)} \longrightarrow \boxed{\text{Sys}} \longrightarrow \begin{matrix} x(-(t-t_0)) \\ x(-t+t_0) \end{matrix}$$

T.V System $\neq y(t-t_0)$

Problems of Time Variant & Time Invariant System

① $y(t) = x(2t)$

$x(t) \longrightarrow x(2t) = y(t)$
 $x(t-t_0) \longrightarrow x(2t-t_0) \text{---①}$

$x(t) \xrightarrow{\text{Sys}} x(2t)$
 $x(t-t_0) \longrightarrow x(2(t-t_0))$
 $\quad \quad \quad x(2t-2t_0)$
T.V $\neq x(2t-t_0)$

② $y(t) = x(t^2)$

$x(t) \longrightarrow x(t^2) = y(t)$
 $x(t-t_0) \longrightarrow x(t^2-t_0) = y(t-t_0) \text{---①}$

$x(t) \xrightarrow{\text{Sys}} x(t^2)$
 $x(t-t_0) \longrightarrow x((t-t_0)^2)$
 $\quad \quad \quad x(t^2 + t_0^2 - 2tt_0)$
T.V $\neq x(t^2-t_0)$
 System

③ $y(t) = x(t+1)$

$x(t) \longrightarrow x(t+1)$
 $x(t-t_0) \longrightarrow x(t+1-t_0) \text{---①}$

$x(t) \xrightarrow{\text{Sys}} x(t+1)$
 $x(t-t_0) \longrightarrow x(t-t_0+1)$
 $\quad \quad \quad = x(t+1-t_0)$
T.I System

④ $y(t) = x^2(t)$

$x(t) \longrightarrow x^2(t) = y(t)$
 $x(t-t_0) \longrightarrow x^2(t-t_0) = y(t-t_0) \text{---①}$

$x(t) \xrightarrow{\text{Sys}} x^2(t)$
 $x(t-t_0) \longrightarrow x^2(t-t_0)$
T. Invariant
 System

⑤ $y(t) = e^{t \cdot x(t)}$

$x(t) \longrightarrow e^{t \cdot x(t)}$
 $x(t-t_0) \longrightarrow e^{(t-t_0) \cdot x(t-t_0)} \text{---①}$

$x(t) \xrightarrow{\text{Sys}} e^{t \cdot x(t)}$
 $x(t-t_0) \longrightarrow e^{t \cdot x(t-t_0)}$
T.V $\neq e^{(t-t_0) \cdot x(t-t_0)}$

⑥ $y(t) = e^{x(t)}$

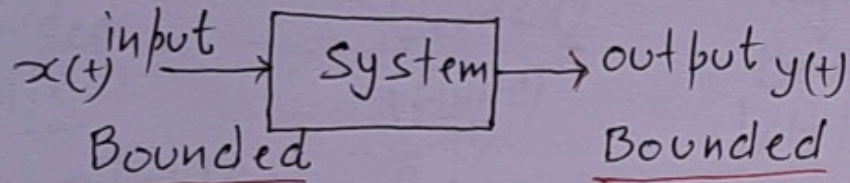
$x(t) \longrightarrow e^{x(t)}$
 $x(t-t_0) \longrightarrow e^{x(t-t_0)} \text{---①}$

$x(t) \xrightarrow{\text{Sys}} e^{x(t)}$
 $x(t-t_0) \longrightarrow e^{x(t-t_0)}$
T.I ✓ Same

Note:- if t is multiplied with $x(t)$, $\frac{dx}{dt}$ it is T.V.
 otherwise Time invariant.

Stable & Unstable system

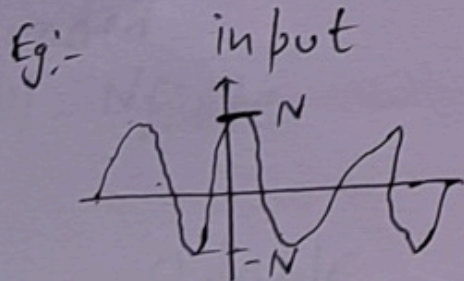
Stable system



Stable → Bounded input
Bounded output
(BIBO stable)

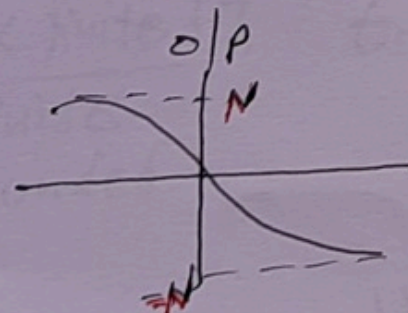
$$|x(t)| \leq N \rightarrow \text{finite for all } t$$

$$|y(t)| \leq N \rightarrow \text{finite for all time}$$



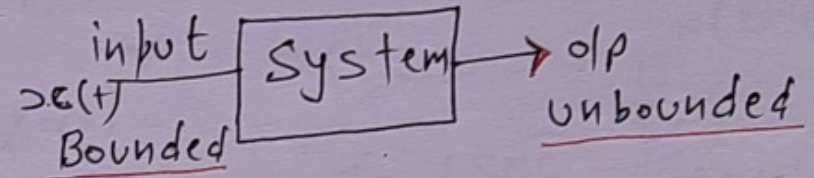
$$|x(t)| \leq N \text{ finite}$$

$$\left. \begin{array}{l} x(t) > -N \\ x(t) \leq N \end{array} \right\} \text{Bounded}$$



$$|y(t)| \leq N \rightarrow \text{finite Bounded o/p}$$

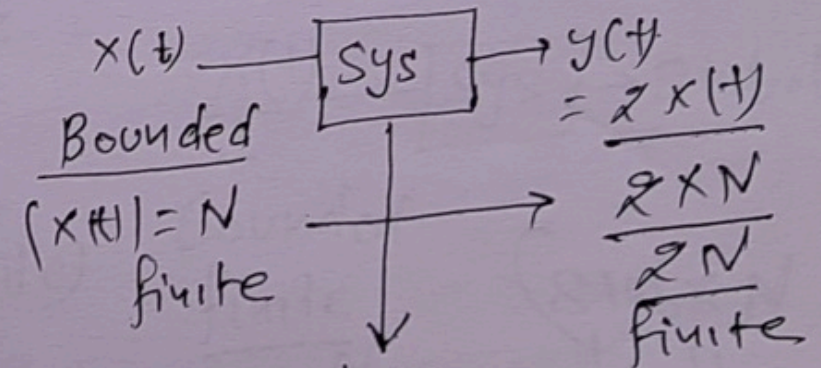
Unstable system



$$|x(t)| \leq N \rightarrow \text{finite}$$

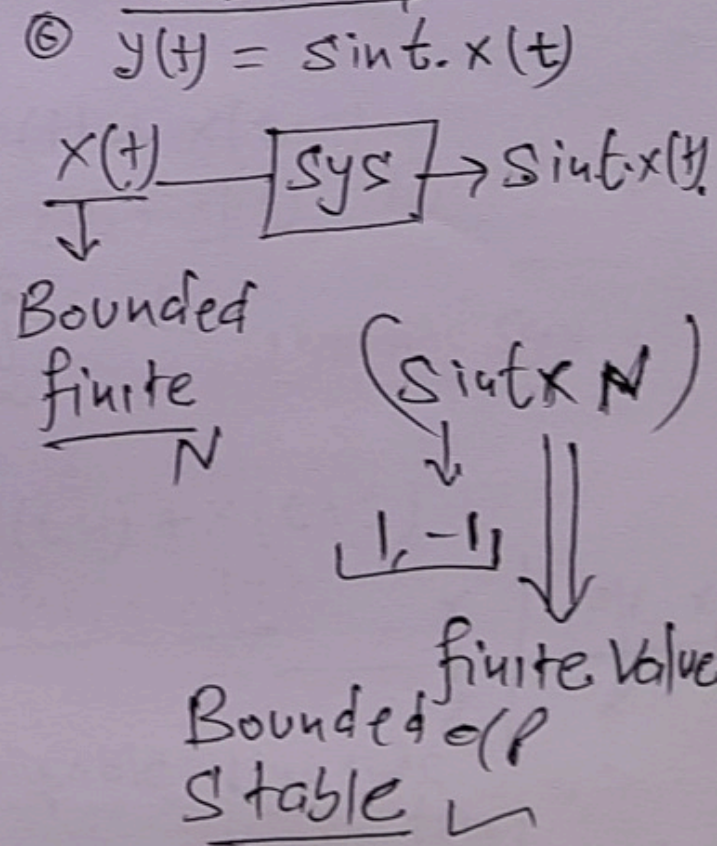
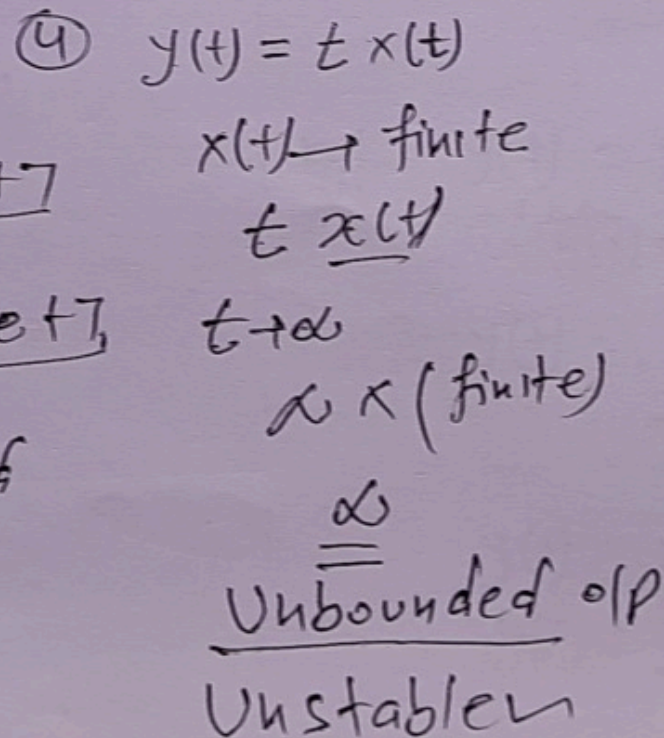
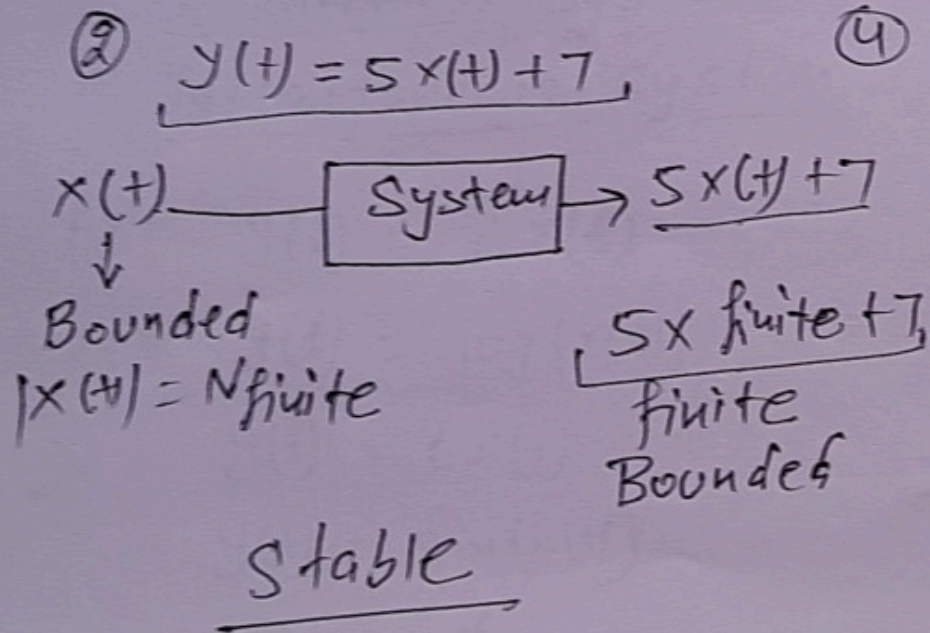
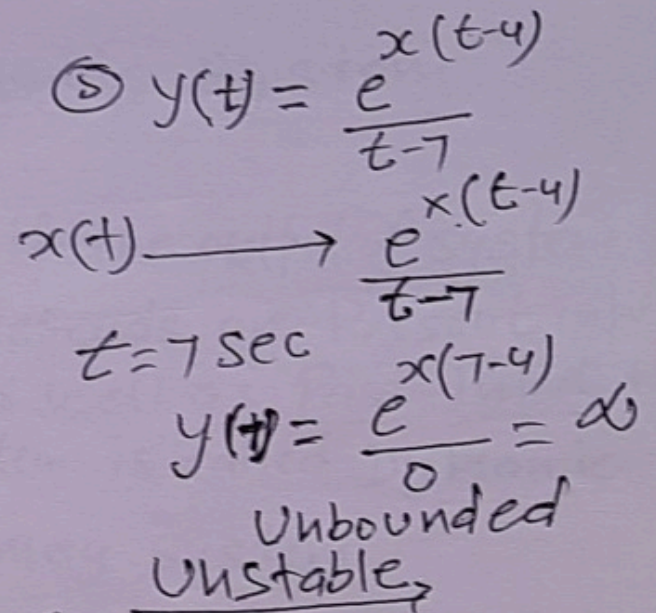
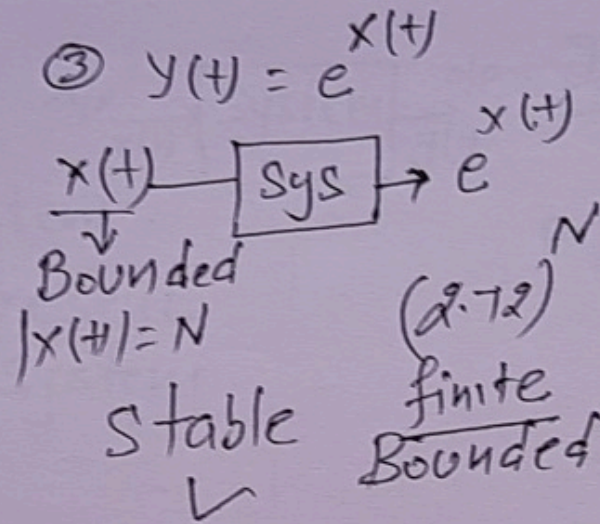
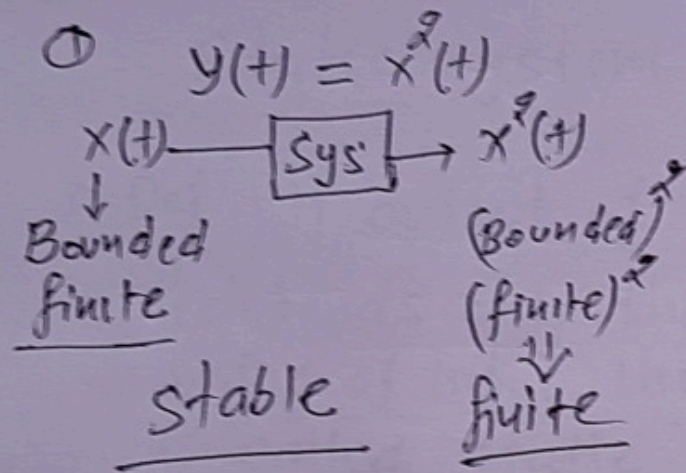
$$|y(t)| = \infty \text{ not bounded}$$

Eg:- $y(t) = 2x(t)$



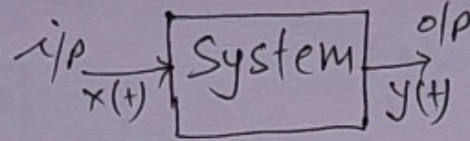
Stable system

Problem of Stable & Unstable system



Static & Dynamic System

Static system



* Def:- if output of system depends only the Present Value of input, then system is called static

* Memoryless system

* Instantaneous system

Eg:- $y(t) = A x(t)$

$$y(t) = \log(x(t))$$

$$y(t) = t x(t)$$

$$y(t) = \sin(x(t))$$

$$y(t) = e^t x(t)$$

i/p, o/p $\rightarrow$ argument (t)

Dynamic System

Def:- if the output of system depends on Present input as well as Past input, then system is called Dynamic

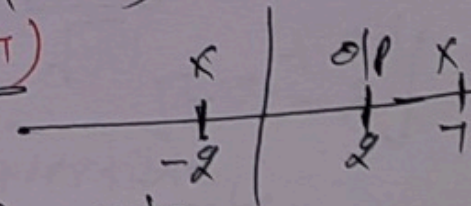
* Memory system

Eg:- $y(t) = x(t) + x(t-1)$
^{1 Sec}
 $y(1) = x(1) + x(-6)$ Dynamic Sys

$$y(t) = \int_{-\infty}^t x(t) \cdot dt$$
 Dynamic Sys

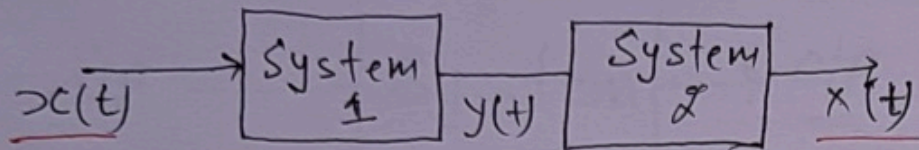
$$y(t) = x(t-4) + x(t+5)$$

$$y(2) = x(-2) + x(7)$$

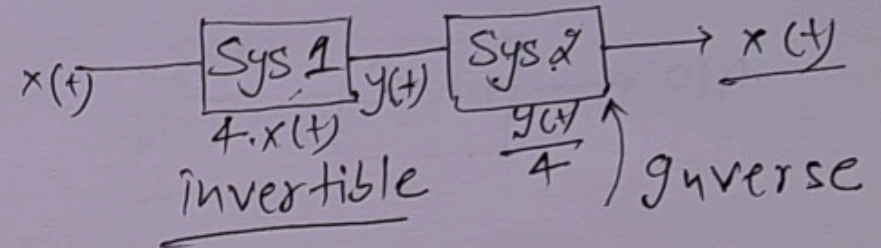


Non Realizable Dynamic system

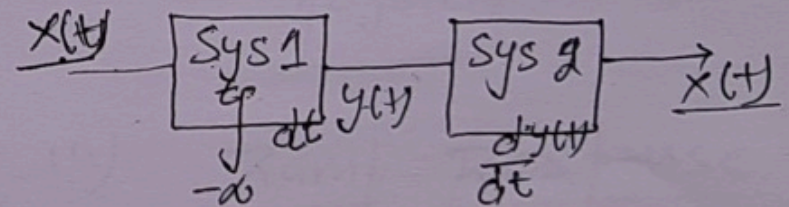
Invertible & Non-invertible System



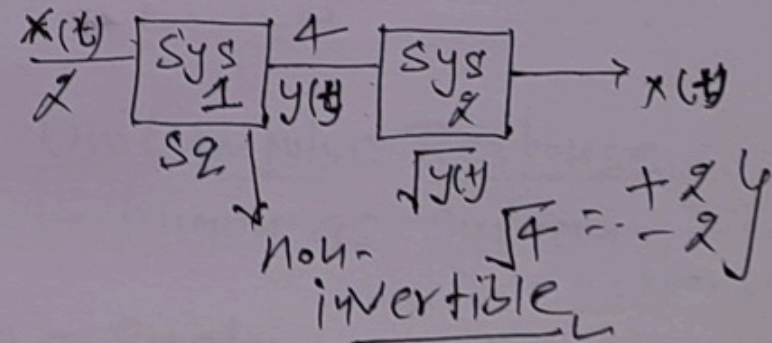
Eg:- $y(t) = 4x(t)$



$$y(t) = \int_{-\infty}^t x(t) \cdot dt$$



$$y(t) = x^2(t)$$



* System 2 performs inverse job of System 1. Such system is called inverse system

* if it is possible to define inverse system then system 1 is called Invertible system

* if inverse system is not defined

then system 1 is called Non-invertible system.

Note:- ① if input multiply with time

② $\frac{dx}{dt} \propto \text{const. } x(t)$

Non-Invertible